

Finding the Hypotenuse

Pythagoras' Theorem. A short guide to walk beside you — read as much or as little as helps.
Connection before curriculum. Always.

What this lesson is about

Every right-angled triangle has one longest side, called the **hypotenuse**. It always sits opposite the right angle. In this lesson you build a clear, step-by-step way to work out how long the hypotenuse is when you already know the two shorter sides. The question you are exploring is: *how can I recognise the hypotenuse and build a correct step-by-step solution for its length?*

What you are learning to notice

- How to find the right angle, and the side that lies opposite it.
- Why we work with the **squares** on the sides, not the side lengths themselves.
- How the theorem $a^2 + b^2 = c^2$ lets you build the length one true line at a time.
- That the last step — the square root — turns an area back into a length.

Moving through it, one step at a time

The builder in the lesson walks through six steps. You do not have to rush them, and you can go back a step whenever you want to. In your own words, the steps are: find the hypotenuse; pick out the two shorter side lengths; write the theorem for this triangle; square each shorter side; add those two results; and finally take the square root to find the length. Each step is one true line of working that follows from the line above it.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (square numbers, square roots, or spotting the hypotenuse), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace. Take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, look for the right angle, find the side opposite it, and start to build.

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